

Deep Learning Lab — Exercise 3 Report

SS2026

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Q 1.1 — kNN Classification: DINOv2 vs. OpenCLIP

DINOv2 performs better, reaching 89.1% at $k=5$, while OpenCLIP peaks at 82.0% at $k=7$. DINOv2 was trained to understand visual details in images, so its features are better at telling apart images that look similar. OpenCLIP was trained on image–text pairs, which helps it understand categories by name but is less useful when you need to spot fine visual differences, as required in SPair-71K.

Q 1.2 — Zero-Shot vs. kNN Classification

Zero-shot CLIP (**81.4%**) is slightly better than OpenCLIP kNN at $k=5$ (80.8%), but both are well below DINOv2 kNN (89.1%). Just using the text prompt "a photo of <category>" gives better results than using real labelled images from the same CLIP model. This shows that CLIP's image and text embeddings are well aligned. Still, DINOv2 wins overall because visual similarity matters more than text matching for this task.

Q 2.1 — PCA Dimensionality and Colour Consistency

	Full	PCA-256	PCA-128	PCA-64	PCA-32	PCA-16
DINO	47.9	48.5	48.3	45.1	39.9	28.6
CLIP	31.7	32.6	32.5	30.4	26.9	25.2

Accuracy stays roughly the same down to PCA-128 and starts dropping noticeably at **PCA-32** for DINO (−8.3 pp drop). PCA-256 and PCA-128 are slightly better than the full features, meaning the removed components were mostly noise. In the PCA-6 colour visualisation, similar body parts across different animals (e.g. cat legs and dog legs) get similar colours. This shows that the features capture meaningful part-level structure, not just object identity.

Q 3.2 — Flow Matching vs. Regression Baseline

Model	Seen (mean)	Unseen (mean)	Unseen "111"
Regression	100.0%	86.7%	33.3%
Flow matching	100.0%	100.0%	100.0%

Both models handle seen attribute combinations perfectly. The difference shows up on **unseen** combinations: regression fails on the "111" case (all three attributes at once), scoring only 33.3%, while flow matching gets 100% on all unseen cases. This is because flow matching learns from a gradually changing image during training, which helps it understand how attributes combine. Regression just learns an average image per condition and cannot generalise well to new combinations.